

Monday, 5th Jan

Analog to Digital Converter (ADC)

1) Output Signal

Analog (Continuous)

- A continuous voltage wave which changes from positive to negative over time (Sine Wave)

Digital (Discrete)

- A steady Voltage (Flat line)
- "High" Dc output signals 1
- "Low" Dc output signals 0

2) Sampling

- measuring the amplitude of continuous analog signal at regular intervals
- Computers cannot store infinite amounts of continuous data, so we use sampling and quantization to turn it into discrete values.

3) Quantization

- Process of mapping large sets of input values to smaller sets of discrete outputs

In real life we use these quantization to convert continuous voltage range to discrete values.

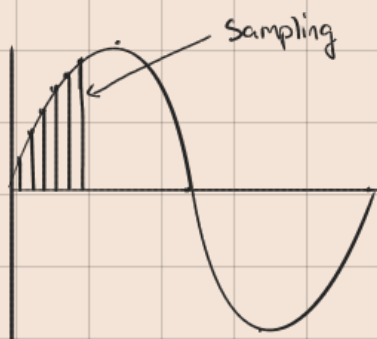
we have a voltage coming from an AC source. the varying input can be quantized by assigning values to distinct voltages

for example 0-1 volts as 0 (In binary logic)

2-5 volts as 1

1-2 volts as forbidden zone to ensure accuracy

This removes the small amounts of unwanted noise and still keeps the wanted varying voltage. Even if the voltage falls from 5 volts to 4.5 volts it will still be considered as on



Analog Input



Quantized Image

4) Digital Signal Outputs

- Result of Sampling and Quantization
- 1's and 0's representing the original analog signal

Number Systems

Decimal (0-9)

- Base (10)

Binary (0-1)

- Base (2)

Hexadecimal (0-15)

- Base (16)

Octal (0-7)

- Base (8)

Decimal	Binary				Hexa	Octal
0	0	0	0	0	0	0
1	0	0	0	1	1	1
2	0	0	1	0	2	2
3	0	0	1	1	3	3
4	0	1	0	0	4	4
5	0	1	0	1	5	5
6	0	1	1	0	6	6
7	0	1	1	1	7	7
8	1	0	0	0	8	10
9	1	0	0	1	9	11
10	1	0	1	0	A	12
11	1	0	1	1	B	13
12	1	1	0	0	C	14
13	1	1	0	1	D	15
14	1	1	1	0	E	16
15	1	1	1	1	F	17

Wednesday, 7th Jan

Binary Addition

$$1 + 1 \rightarrow 0 \text{ (1 carry)}$$

$$1 + 0 \rightarrow 1$$

$$0 + 1 \rightarrow 1$$

$$0 + 0 \rightarrow 0$$

$$1 + 1 + 1 \rightarrow 1 \text{ (1 carry)}$$

$$\begin{array}{r} 10101 \\ + 1011 \\ \hline 10100 \end{array}$$

$$\begin{array}{r} 10011 \\ + 1111 \\ \hline 10010 \end{array}$$

Binary Subtraction

$$12 - 8 = ?$$

$$12 = 00001100$$

$$8 = 00001000$$

1) Perform 2's complement on 8

$$8 = 00001000$$

$$\begin{array}{r} 1111 \\ 1110111 \end{array}$$

$$\begin{array}{r} + 1 \\ \hline \end{array}$$

$$1111000$$

$$= 1111000$$

2) Add the two numbers

$$\begin{array}{r} 00001100 \\ + 11110000 \\ \hline \end{array} \quad (12)$$

$$+ 11110000 \text{ (2's complement of 8)}$$

$$\begin{array}{r} 100000100 \end{array}$$

Ignore MSB

$$\begin{array}{r} 128 \quad 64 \quad 32 \quad 16 \quad 8 \quad 4 \quad 2 \quad 1 \\ 0 \quad 0 \quad 0 \quad 0 \quad 0 \quad 1 \quad 0 \quad 0 \end{array}$$

$$12 - 8 = 4$$

Binary Subtraction (Negative Case)

$$6 - 10 = ?$$

$$6 = 0000110$$

$$10 = 00001010$$

1) perform 2's Complement on 10

$$\begin{array}{r} 00001010 \\ 11110101 \\ + \quad \quad \quad 1 \\ \hline 11110110 \end{array}$$

Invert

2) Add the two numbers

$$\begin{array}{r} 0000110 \quad (6) \\ + 11110110 \quad (2's \text{ complement of } 10) \\ \hline 1111100 \end{array}$$

negative number

3) Apply 2's complement on the answer and make the MSB negative / on

$$\begin{array}{r} 1111100 \\ 0000111 \\ + \quad \quad \quad 1 \\ \hline 0000100 \end{array}$$

Invert

0000100
↓
10000100

Binary multiplication

$$15 \times 5 = ?$$

$$15 = \text{00001111}$$

$$5 = \text{0000101}$$

$$0 \times 0 = 0$$

$$0 \times 1 = 0$$

$$1 \times 0 = 0$$

$$1 \times 1 = 1$$

1) Ignore insignificant zeroes and multiply

$$\begin{array}{r}
 1111 \\
 \times 101 \\
 \hline
 1111 \\
 0000 \\
 + 1111 \\
 \hline
 1001011
 \end{array}$$

$$\begin{array}{ccccccc}
 64 & 32 & 16 & 8 & 4 & 2 & 1 \\
 1 & 0 & 0 & 1 & 0 & 1 & 1 \\
 \hline
 & & & & & & = 75
 \end{array}$$

Binary Division

$$25 \div 5 = ?$$

$$25 = 11001$$

$$5 = 101$$

$$\text{Answer} = 00101$$

$$\begin{array}{r}
 00101 \\
 101 \overline{) 11001} \\
 \underline{- 101} \\
 00101 \\
 \underline{- 101} \\
 000
 \end{array}$$

$$40 \div 4 = ?$$

$$40 = \cancel{00} 101000$$

$$4 = \cancel{0000} 0100$$

$$\text{Ans} = 1010$$

$$\begin{array}{r}
 100 \overline{) 001010} \\
 \underline{100} \\
 00100 \\
 \underline{100} \\
 00000
 \end{array}$$

Overflow in Signed Arithmetic

- Overflow can only occur in Addition and subtraction

Overflow in 1's complement

$$+5 = 1 \overset{1}{0} \overset{1}{1} \overset{1}{0} 1$$

$$-4 = + \quad 1 \quad 0 \quad 1 \quad 1 \quad (1's \text{ complement negative})$$

$$\begin{array}{r}
 \overset{\text{overflow}}{\circledast} 1 \quad 0 \quad 0 \quad 0 \quad 0 \\
 \quad \quad \quad 0 \quad 0 \quad 0 \quad 0 \quad 1
 \end{array}$$

$$\text{Ans} = 0 \quad 0 \quad 0 \quad 1$$

∴ negative numbers
 can be represented in
 two ways, 1's complement
 and 2's complement

Overflow in 2's Complement

- occurs when adding two numbers with the same sign but result has opposite sign.
- **Positive overflow**: two positive numbers produce a negative
- **Negative overflow**: two negative numbers produce a positive

$$\begin{array}{r} +7 = \quad \overset{1}{0} \overset{1}{1} \overset{1}{1} \overset{1}{1} \\ +3 = + \quad \underline{0 \ 0 \ 1 \ 1} \\ \hline 1 \ 0 \ 1 \ 0 \end{array}$$

negative

$$\textcircled{1} \ 0 \ 1 \ 0 = -6$$

$$-5 = \quad \underline{1 \ 1 \ 0 \ 1 \ 1} \quad (2's \ complement)$$

$$-4 = + \quad \underline{1 \ 1 \ 0 \ 0} \quad (2's \ complement)$$

$$\begin{array}{r} \text{lost} \leftarrow \textcircled{1} \ 0 \ 1 \ 1 \ 1 \end{array}$$

positive

$$\textcircled{0} \ 1 \ 1 \ 1 = 12$$

$$+5 = \quad \overset{1}{1} \ 0 \ 1 \ 0 \ 1$$

$$-4 = + \quad \underline{1 \ 1 \ 0 \ 0} \quad (2's \ complement)$$

$$\begin{array}{r} \text{lost} \leftarrow \textcircled{1} \ 0 \ 0 \ 0 \ 1 \end{array}$$

$$\text{answer} = \quad 0 \ 0 \ 0 \ 1$$

Solution for overflow

- 1) Increase the number of bits used
- 2) Detect overflow and generate a flag
- 3) Saturation Arithmetic
- 4) Check before Addition

Monday, 12th Jan

AND Gate



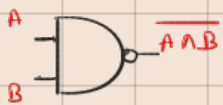
A	B	$A \text{ AND } B$
0	0	0
0	1	0
1	0	0
1	1	1

OR Gate



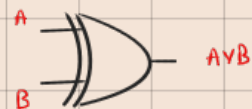
A	B	$A \text{ OR } B$
0	0	0
0	1	1
1	0	1
1	1	1

NAND Gate



A	B	$\neg(A \text{ AND } B)$
0	0	1
0	1	1
1	0	1
1	1	0

XOR Gate



A	B	$A'B + AB'$
0	0	0
0	1	1
1	0	1
1	1	0

NOR Gate



A	B	$\neg(A \text{ OR } B)$
0	0	1
0	1	0
1	0	0
1	1	0

Application of Logic Gates

1) Used In Circuits

- thermostats
- push button locks
- automatic watering Systems
- light activated burglar alarms

2) Computer and Processors

- Designing CPU's and GPU's

3) Arithmetic Circuits

- Adders, Subtrahors, Multipliers

4) Control Systems

- Automated devices
- Decision making Circuits

5) Memory Devices

- Flip Flops

Boolean Algebra

Basic Theorems and Laws of Boolean Algebra

Part-A	Part-B
$A+0=A$	$A.0=0$
$A+1=1$	$A.1=A$
$A+A=A$ (Idempotence law)	$A.A=A$ (Idempotence law)
$A+\bar{A}=1$	$A.\bar{A}=0$
$\overline{\overline{A}}=A$ (double inversion law)	--
Commutative law: $A+B=B+A$	$A.B=B.A$
Associative law: $A+(B+C)=(A+B)+C$	$A(B.C)=(A.B)C$
Distributive law: $A.(B+C)=AB+AC$	$A+BC=(A+B).(A+C)$
Absorption law: $A+AB=A$	$A(A+B)=A$
DeMorgan Theorem: $\overline{(A+B)}=\bar{A}.\bar{B}$	$\overline{(A.B)}=\bar{A}+\bar{B}$
Redundant Literal Rule: $A+\bar{A}.B=A+B$	$A.(\bar{A}+B)=AB$

6×7

$$6 = \begin{array}{r} 0110 \\ \times \quad 111 \\ \hline \end{array}$$

7: ~~0~~ 1 1 1 1 1 1 0

1	1	1	1	1	0
		1	1	0	X
+	1	1	0	X	X
1	0	1	0	1	0

1	1	0	x
---	---	---	---

+	1	1	0	x	x
---	---	---	---	---	---

1	0	1	0	1	0
---	---	---	---	---	---

	32	16	8	4	2	1		
=	1	0	1	0	1	0		= 42

=	1	0	1	0	1	0		= 42
---	---	---	---	---	---	---	--	------

1	1	1	1
x	1	0	1

x	1	0	1
---	---	---	---

1	1	1	1	1	1	1
---	---	---	---	---	---	---

0	0	0	0	X
---	---	---	---	---

+	1	1	1	1	x	x
---	---	---	---	---	---	---

1	0	0	1	0	1	1
---	---	---	---	---	---	---

=	1	1	0	1	0	1	1	1
---	---	---	---	---	---	---	---	---

$$50 \div 5$$

$$50 = \quad \text{0 0 1 1 0 0 1 0}$$

$$5 = \quad \text{0 1 0 1}$$

$$\begin{array}{r}
 101 \overline{) 001010} \\
 \underline{110010} \\
 001010 \\
 \underline{0110} \\
 0000
 \end{array}$$

$$2001010$$

$$10110$$

$$\begin{array}{r}
 101 \\
 010 \\
 \text{+ } 1 \\
 011
 \end{array}$$

32	16	8	4	2	1
1	0	1	1	0	1

$$\begin{array}{r}
 \begin{array}{ccccccc}
 & 0 & 0 & 1 & 1 & 1 & 1 \\
 11 \overline{) 001101} & & & & & & \\
 \underline{11} & & & & & & \\
 & 0 & 1 & 1 & 0 & 1 & \\
 & \underline{11} & & & & & \\
 & & 0 & 1 & & & \\
 & & \underline{11} & & & & \\
 & & & 1 & 0 & & \\
 & & & \underline{11} & & & \\
 & & & & 0 & 1 & 1 \\
 & & & & \underline{11} & & \\
 & & & & & 0 & 0
 \end{array}
 \end{array}$$

$$\text{ans} = 1111$$

$$\begin{array}{r}
 \begin{array}{ccccccc}
 & 0 & 0 & 1 & 1 & 0 & 0 \\
 11 \overline{) 001100} & & & & & & \\
 \underline{11} & & & & & & \\
 & 0 & 1 & 1 & 0 & 0 & \\
 & & \underline{11} & & & & \\
 & & & 0 & 0 & &
 \end{array}
 \end{array}$$

$$\begin{array}{ccccccc}
 32 & 16 & 8 & 4 & 2 & 1 & \\
 1 & 0 & 0 & 1 & 0 & 0 & = 36
 \end{array}$$

$$= 1100$$

$$\begin{array}{r}
 0011 \\
 011 \overline{) 100100} \\
 \underline{+ 101} \\
 \textcircled{1} 001
 \end{array}$$

$$\begin{array}{r}
 011 \\
 100 \\
 101
 \end{array}$$

$$(A+B) \cdot (A+C)$$

$$A + AC + AB + BC$$

$$A + AC + AB + BC$$

$$A + BC$$

$$A + AC + BC$$

$$\underline{A + AC} + AB + BC$$

$$\underline{A + AB} + BC$$

$$A + BC$$

$$A = 1$$

$$A = 0$$

$$(1 \times 1)$$

$$= 1$$

$$0 \times 0$$

$$= 0$$

$$A + A = A$$

Idempotence law